

HACKATHON 2026 · USE CASE #12 · INSURANCE / SIU

Fraud Pattern Radar

catch the whole ring — and prove it.

Explainable, real-time claim-fraud scoring with organised-ring detection. Every flag traces to a named rule — the **maths** decides; AI just helps at the edges.

Team Overwatch

Deterministic-first

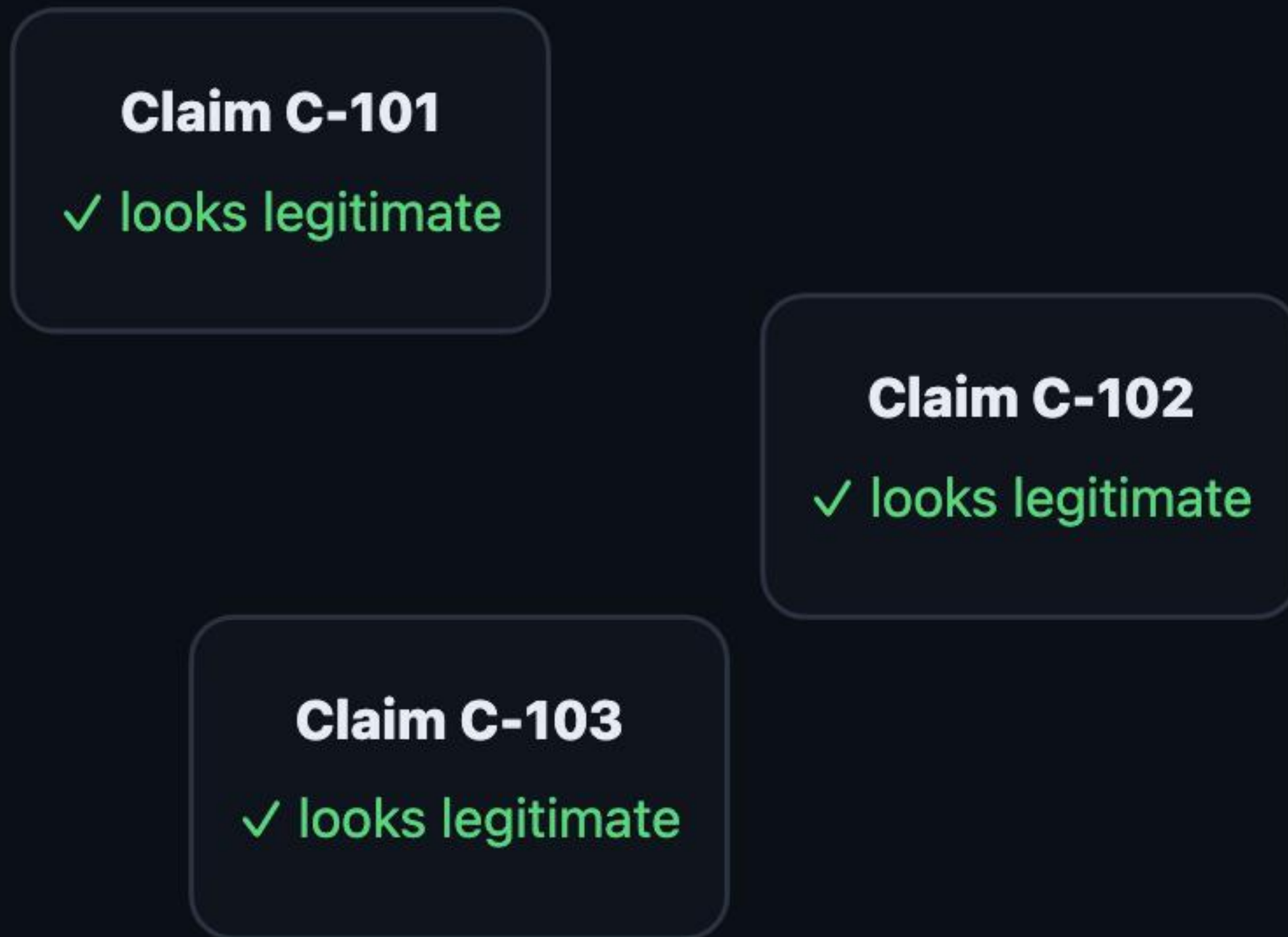
Network-aware

Runs offline

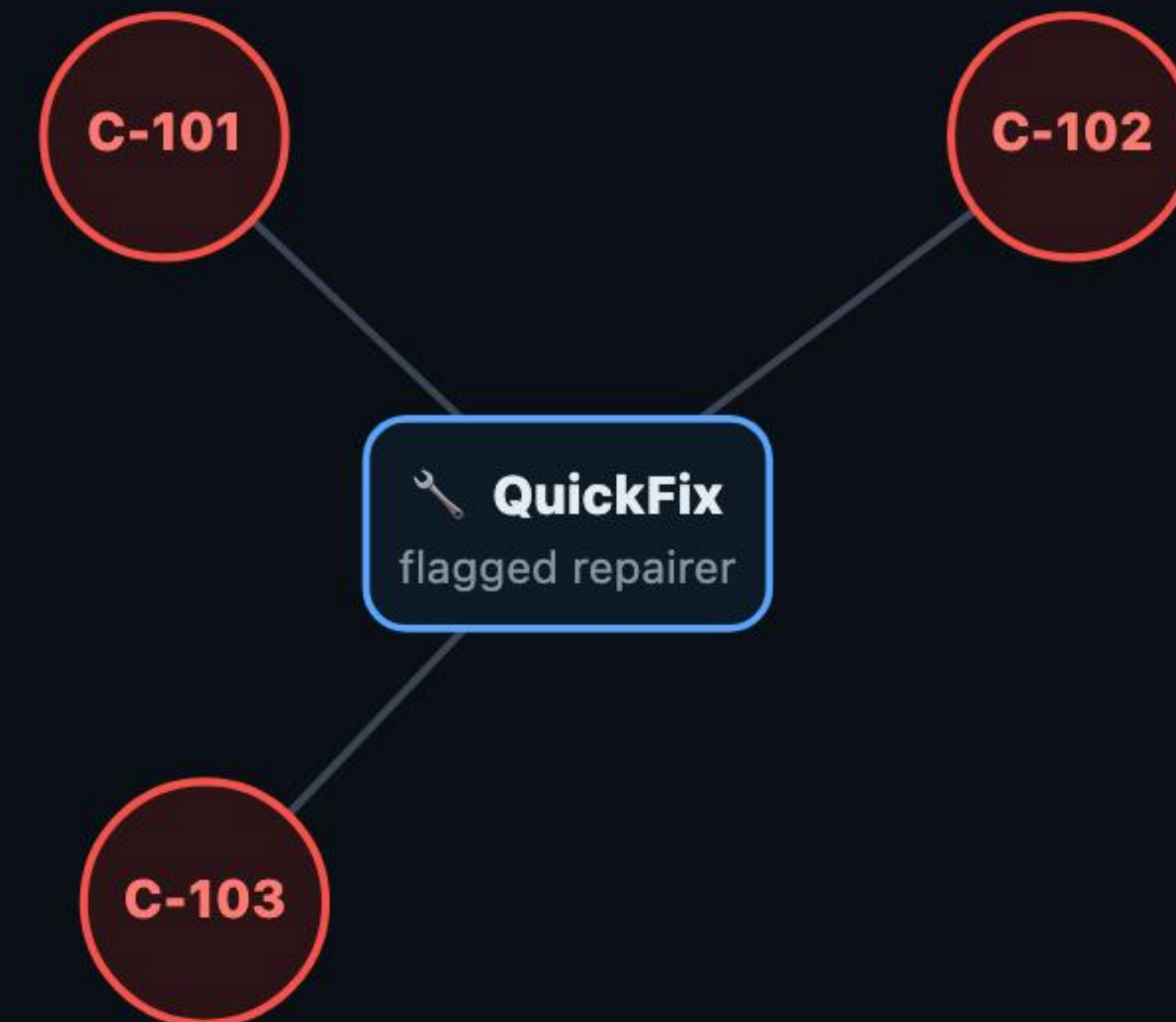
THE PROBLEM

Fraud hides **between** claims, not inside one

One claim at a time — each looks clean



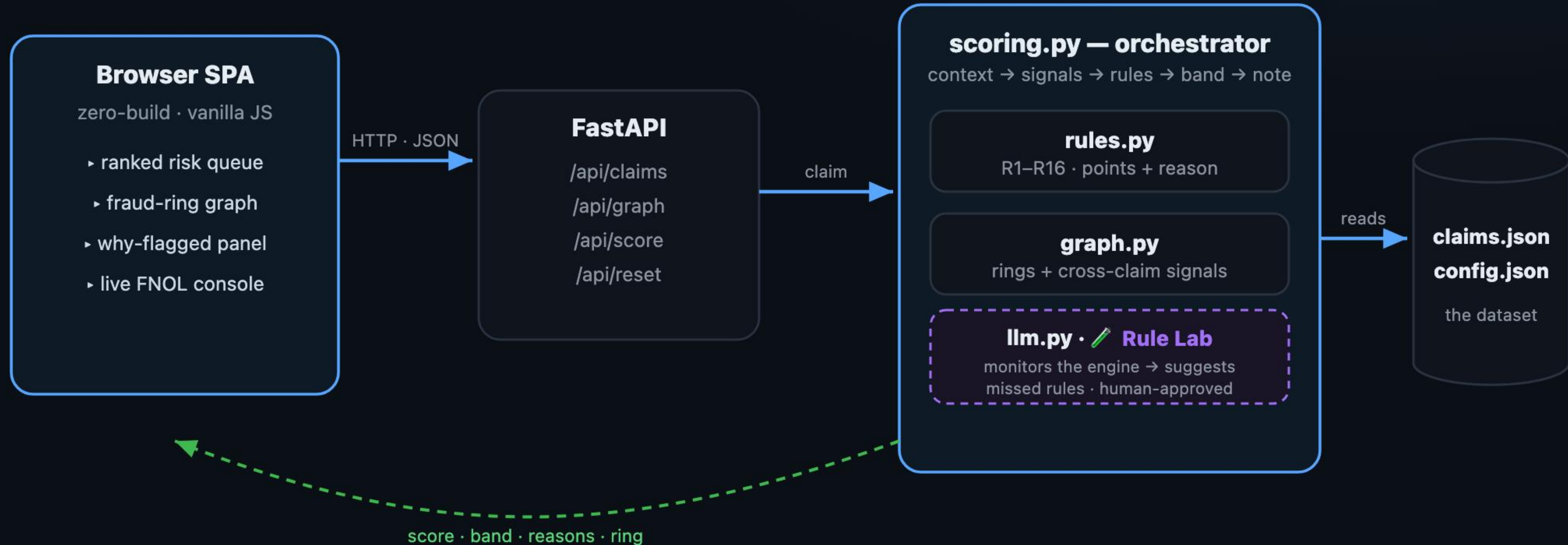
Across the book — the ring appears



Same three claims, three different policyholders. Reviewed alone, each passes. Only the **network** reveals they all funnel through one flagged repair shop.

ARCHITECTURE

Small, layered, deterministic-core — dataset-driven



Scoring is **100% reproducible** — the AI reads messy claims, writes notes / referrals, and **monitors the engine to suggest missed rules a human approves** (📝 Rule Lab), all on **local Ollama**. Swap `claims.json` for real data, **zero code changes**.

HOW IT WORKS

One claim → a routed, explained decision



Worked example — the walk-in claim, points accumulating to a score:

R1 +15	R12 +15	R13 +10	R8 +12	R7 +10	R6 +5	R11 +7	= 74	HIGH → SIU
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The score is always the **exact sum of fired-rule points** — additive, reproducible, and auditable line by line.

THE ENGINE

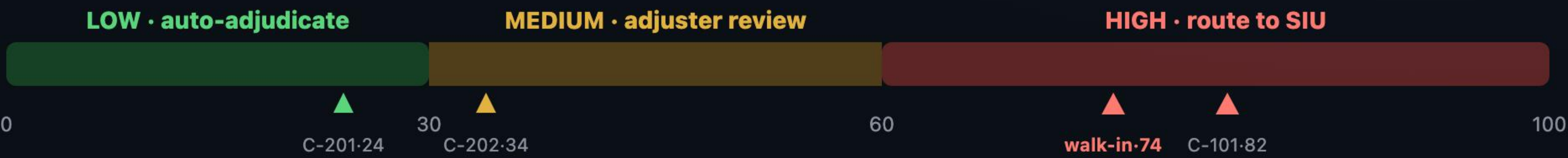
16 rules across 6 fraud categories

R1 Just-in-time cover +15	R2 Reinstated pre-loss +10	R3 Address mismatch +8	R4 Late report +8
R5 Serial claimant +12	R6 Weak-witness timing +5	R7 Above 90th pct amount +10	R8 Threshold gaming +12
R9 Fabricated invoices +8	R10 Fast cash, no inspect +10	R11 No police report +7	R12 Shared-identifier ring +15
R13 Flagged repairer +10	R14 Doc inconsistency +10	R15 Duplicate photo +15	R16 EXIF mismatch +8

Colour = category · number = points. Every rule is tested positive, negative, **and at its exact boundary.**

SCORE BANDS → ACTION

The score drives a concrete routing decision

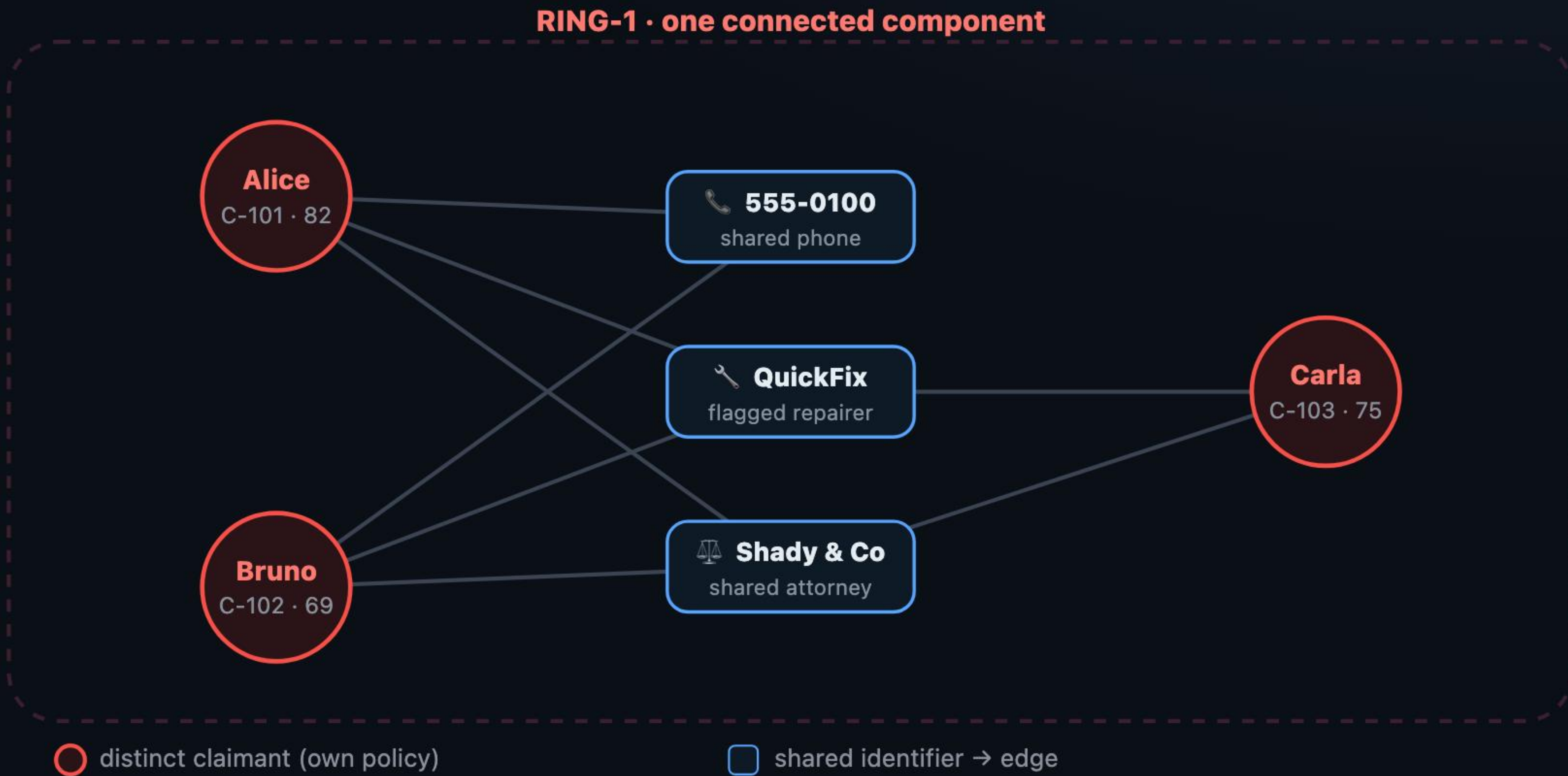


BAND	SCORE	ROUTING ACTION
LOW	0 – 29	Straight-through / auto-adjudicate
MEDIUM	30 – 59	Adjuster review with reason codes
HIGH	≥ 60	Route to SIU queue — top of the investigator list

Straight from the #12 / #48 rule card. This one number powers both the **ranked queue** and the **auto-routing**.

THE CENTREPIECE

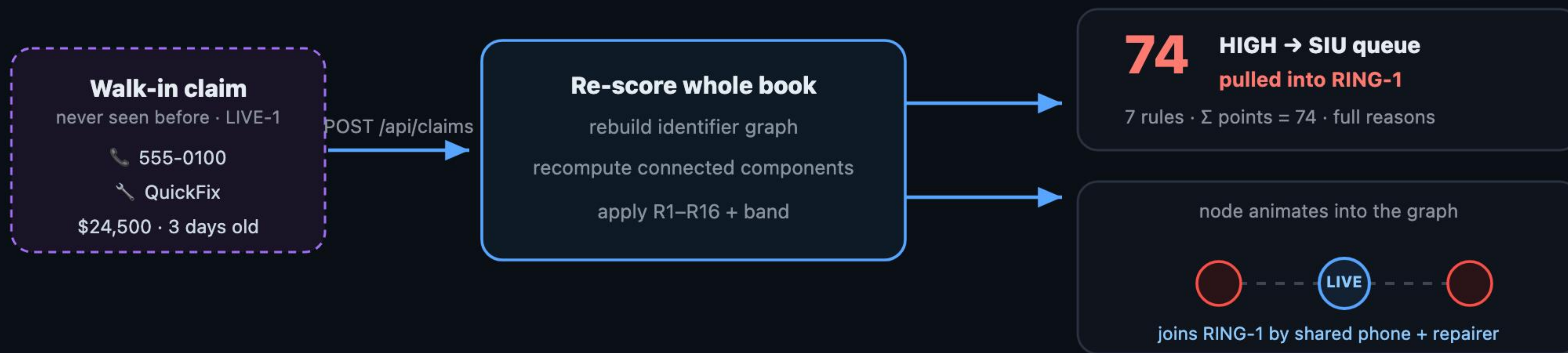
Fraud-ring detection via network analysis



Index every identifier; a value spanning ≥ 2 **distinct claimants** becomes an edge. Rings are the **connected components**. Same-claimant reuse is *not* a ring — that's the serial-claimant rule (R5). In the dashboard this renders **live as an entity graph**: every shared detail (📞 / 🔧 / ⚖️) is its own node, so a box wired to several claimants *is* the ring — you see it, you don't decode it.

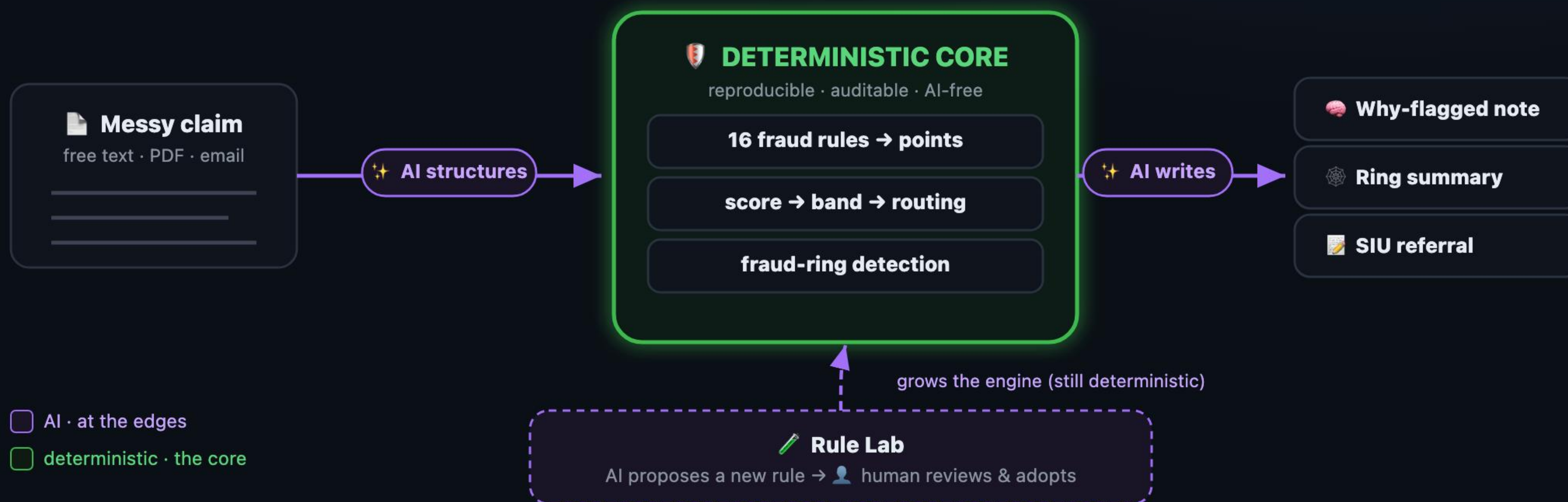
LIVE AT INTAKE

A brand-new claim joins the ring in real time



Detection is **dynamic, not precomputed** — the ring genuinely re-links the moment the claim lands. In the dashboard, click + Score claim at intake → ● Ring member to show it live.

AI **wraps** the engine — it never decides the verdict



- ☐ AI · at the edges
- ☒ deterministic · the core

\$0 to run — local Ollama · no per-token fees · data never leaves your walls

Messy → structured going in, structured → readable coming out, and a human-gated way to grow the rules. The score, band and ring in the middle are **never** AI-decided.

SCOPE COVERAGE

All 6 use cases · all 4 MVP items

REQUIREMENT	DELIVERED BY	
1 · Scoring at FNOL / intake	Live POST /api/score + in-UI console	✓
2 · Anomaly: amount / timing / frequency	R7,R8 · R1,R2,R4,R6 · R5	✓
3 · Entity linking (claimants / providers / addresses)	Shared-identifier graph + R5 / R13	✓
4 · SIU prioritization	Ranked queue + band-driven routing	✓
5 · Explanation layer	Per-rule reasons + narrative + additive score	✓
6 · Fraud-ring detection	Connected-component network analysis	✓
MVP · risk score · why-flagged · duplicate detection · ranked queue	All four, end-to-end	✓✓✓✓

Every row is backed by an automated conformance test — not just a claim in a slide.

QUALITY

Built to be trusted, not just demoed

103

automated checks — all green

16 x 3

rules x positive / negative / boundary

404 · 422

API robustness covered

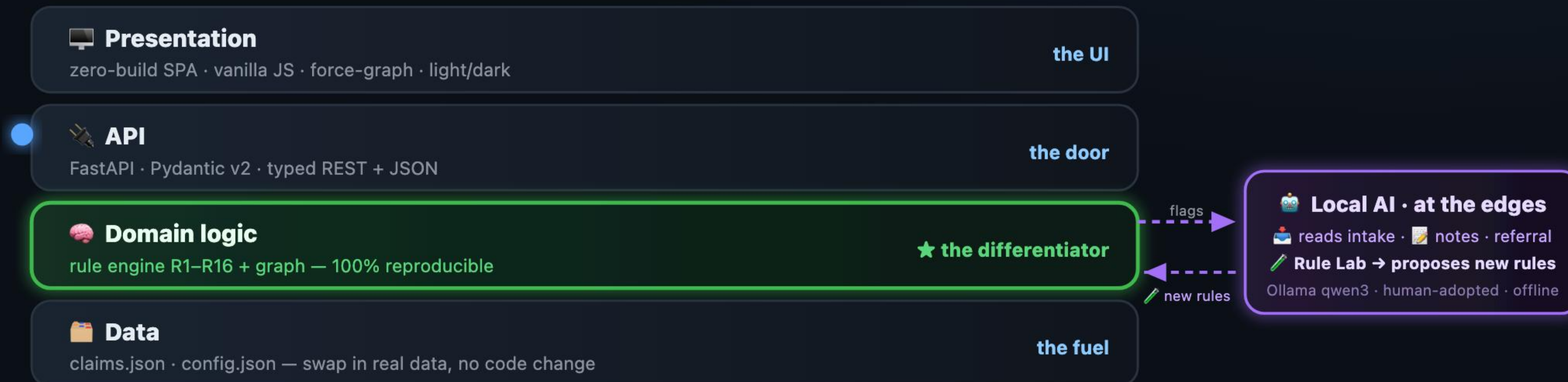
0 FP

clean controls stay clean

- Every rule fires when it should — and **stays silent** when it shouldn't (boundary-exact: 29 vs 30, 59 vs 60).
- Ring logic verified: distinct-claimant requirement, edge de-dup, same-claimant $\neq$ ring.
- Score additivity asserted: $\Sigma \text{ points} == \text{score}$ on every claim.

TECH

Boring where it counts, **sharp** where it matters



A claim falls **down the stack** to the deterministic core. **Local AI** hangs off the side — it reads messy intake, writes the notes & SIU referral, and via  **Rule Lab** even **proposes new rules** for a human to adopt into the core.

ROADMAP · THANK YOU

Team **Overwatch**

Catch the whole ring — and prove it.



▲ you are here



Explainable
every point → a named rule



Network-aware
catches organised rings



Dataset-driven
swap in real data, no code



Offline-safe
no key required to run

Live demo → <http://localhost:8000> · deck → </deck> · **103 / 103** tests green